



MISO DPP 2019 Cycle1 Michigan Area Study Phase 1 Report

January 21, 2021

(FINAL)

**MISO
720 City Center
Drive Carmel
Indiana 46032**
<http://www.misoenergy.org>

Contents

1. Executive Summary	5
1.1. Project List.....	5
2. FERC Order 827 Compliance Review.....	11
3. Model Development and Study Assumptions.....	12
3.1. Base Case Models	12
3.2. Monitored Elements.....	13
3.3. Contingencies.....	13
3.4. Study Methodology.....	13
3.5. Performance Criteria.....	13
4. Thermal Analysis.....	13
4.1. J1248.....	14
4.2. J1310.....	14
4.3. J1320.....	14
4.4. J1394.....	14
4.5. J1430.....	15
4.6. J1472.....	15
5. Voltage Analysis.....	15
6. Stability Analysis.....	15
7. Short Circuit Analysis	15
8. Affected System Impact Study.....	15
9. Deliverability Analysis.....	15
9.1. Introduction.....	15
9.2. Determining the MW Restriction	16
9.3. Deliverability Study Results.....	16
9.4. J1192.....	17
9.5. J1194.....	17
9.6. J1196.....	18
9.7. J1203.....	18
9.8. J1206.....	19
9.9. J1207.....	20
9.10. J1210.....	20
9.11. J1211.....	21
9.12. J1224.....	22
9.13. J1226.....	22
9.14. J1248.....	23
9.15. J1255.....	23
9.16. J1292.....	24
9.17. J1297.....	24

9.18.	J1307.....	25
9.19.	J1310.....	25
9.20.	J1319.....	26
9.21.	J1320.....	27
9.22.	J1322.....	28
9.23.	J1327.....	28
9.24.	J1329.....	29
9.25.	J1331.....	29
9.26.	J1350.....	30
9.27.	J1375.....	30
9.28.	J1379.....	31
9.29.	J1389.....	32
9.30.	J1394.....	32
9.31.	J1399.....	33
9.32.	J1401.....	34
9.33.	J1403.....	35
9.34.	J1430.....	35
9.35.	J1449.....	36
9.36.	J1472.....	37
10.	Shared Network Upgrades Analysis	38
11.	Cost Allocation.....	38
11.1.	Cost Assumptions for Network Upgrades.....	38
11.2.	Cost Allocation Methodology.....	38
12.	Revisions to the Draft report.....	38
	Appendix A – Cost Allocation Summary (CEII)	40
	Appendix B – Network Upgrade Table Per Project (CEII).....	40
	Appendix C – MISO ERIS Analysis (CEII)	40
	Appendix D- MISO Stability Analysis.....	40
	Appendix E – ITC Short Circuit Analysis (CEII).....	40
	Appendix F – PJM Affected Systems Study Report (CEII).....	40
	Appendix G – CE Affected Systems Study Report (CEII)	40
	Appendix H – DTE Affected Systems Study Report (CEII)	40
	Appendix I – WPSC Affected Systems Study Report (CEII).....	40
	Appendix J – MISO Deliverability Analysis (CEII).....	40
	Appendix K– MISO A10-Conditionality Analysis (CEII).....	40
	 Table 1: List of DPP 2019 Cycle 1 Michigan Area Phase 1 Projects	5
	Table 2: Total Cost of Network Upgrades for DPP 2019 Cycle1 Michigan Phase 1 Projects	7
	Table 3: ERIS & NRIS Upgrades (Planning level cost estimates)	9
	Table 4: FERC Order 827 Review Results.....	11

Table 5: DPP 2019 Cycle1 Michigan-Scenarios	12
Table 6: NRIS Results for J1192.....	17
Table 7: NRIS Results for J1194.....	17
Table 8: NRIS Results for J1196.....	18
Table 9: NRIS Results for J1203.....	18
Table 10: NRIS Results for J1206.....	19
Table 11: NRIS Results for J1207.....	20
Table 11: NRIS Results for J1210.....	20
Table 12: NRIS Results for J1211.....	21
Table 13: NRIS Results for J1224.....	22
Table 14: NRIS Results for J1226.....	22
Table 15: NRIS Results for J1248.....	23
Table 16: NRIS Results for J1255.....	23
Table 17: NRIS Results for J1292.....	24
Table 18: NRIS Results for J1297.....	24
Table 19: NRIS Results for J1307.....	25
Table 20: NRIS Results for J1310.....	26
Table 21: NRIS Results for J1319.....	26
Table 22: NRIS Results for J1320.....	27
Table 23: NRIS Results for J1322.....	28
Table 24: NRIS Results for J1327.....	28
Table 25: NRIS Results for J1329.....	29
Table 26: NRIS Results for J1331.....	29
Table 27: NRIS Results for J1350.....	30
Table 28: NRIS Results for J1375.....	30
Table 29: NRIS Results for J1379.....	31
Table 30: NRIS Results for J1389.....	32
Table 31: NRIS Results for J1394.....	32
Table 32: NRIS Results for J1399.....	33
Table 33: NRIS Results for J1401.....	34
Table 34: NRIS Results for J1403.....	35
Table 35: NRIS Results for J1430.....	35
Table 36: NRIS Results for J1449.....	36
Table 37: NRIS Results for J1472.....	37



1. Executive Summary

This report presents the results of a System Impact Study (SIS) performed to evaluate the interconnection of the generators in the DPP 2019 Cycle1 Michigan Area Phase 1 (Michigan Area DPP-2019-Cycle1 Phase 1). The study was performed under the direction of MISO and reviewed by an ad hoc study group. The ad hoc study group was formed to review the study scope, methodology, models and results. The ad hoc study group consisted of representatives from the interconnection customers and the following utility companies:

- ATC
- MPower
- Consumers Energy
- PJM
- NIPSCO
- DTE
- Wolverine
- ITC

1.1. Project List

The original interconnection requests for DPP 2018 Cycle1 Michigan Area had a total of 38 generation projects: J1192, J1194, J1196, J1203, J1206, J1207, J1210, J1211, J1224, J1226, J1248, J1255, J1262, J1292, J1297, J1307, J1310, J1319, J1320, J1322, J1327, J1329, J1331, J1350, J1370, J1375, J1379, J1389, J1394, J1399, J1401, J1403, J1430, J1449 and J1472. Therefore, there are 38 projects generation projects with a combined nameplate rating of 4190.7 MW (ERIS) & 4165.7 MW (NRIS). The detailed list of Michigan Area DPP Phase 1 is shown below in Table 1, and the Phase 1 study was kicked off on July 7, 2020.

Table 1: List of DPP 2019 Cycle 1 Michigan Area Phase 1 Projects

Project #	Fuel	Summer MW	Transmission Owner	POI Name	County	State	Service Type
J1192	Solar	35	METC	Oneida 138kV Substation	Eaton County	MI	NRIS
J1194	Wind	225.2	METC	Murphy 138 kV Substation	Midland County, Saginaw County	MI	NRIS
J1196	Solar	200	ITC	Greenwood - Rapson (Banner) 345 kV Line Tap	Sanilac County	MI	NRIS
J1203	Solar	125	METC	Thetford 345kV Substation	Genesee County	MI	NRIS
J1206	Gas	50	METC	Newton 138kV Substation	Branch County	MI	NRIS
J1207	Gas	50	METC	Newton 138kV Substation	Branch County	MI	NRIS

Project #	Fuel	Summer MW	Transmission Owner	POI Name	County	State	Service Type
J1210	Solar	50	METC	Whittemore - Twining 138kV Line	Iosco County	MI	NRIS
J1211	Solar	50	CE	Vernon - METC Tap 138kV Line	Isabella County	MI	NRIS
J1224	Solar	170	ITC	Majestic - Milan 345 kV Line Tap	Washtenaw County	MI	NRIS
J1226	Solar	200	METC	Morocco 345 kV Substation	Lenawee County	MI	NRIS
J1248	Solar	100	METC	Marshall - Black Stone 138 kV line	Calhoun County	MI	NRIS
J1255	Solar	200	METC	Marquette - Vergennes 138 kV Line Tap	Ionia County	MI	NRIS
J1262	Gas	25	ITC	Waterman 120kV Substation	Wayne County	MI	ERIS
J1292	Wind	198	METC	Thompson Road 138kV Substation	Ionia County	MI	NRIS
J1297	Solar	170	METC	Cornell - Bell Road 138 kV Line	Shiawassee County	MI	NRIS
J1307	Solar	150	METC	Oneida - Battle Creek 345kV Line	Eaton County	MI	NRIS
J1310	Solar	125	METC	Blackstone - Marshall 138 kV Line	Jackson County	MI	NRIS
J1319	Wind	117.5	ITC	Rapson 120kV Substation	Huron County	MI	NRIS
J1320	Solar	150	METC	Coldwater - Batavia 138kV Line	Branch County	MI	NRIS
J1322	Solar	200	METC	Battle Creek - Oneida 345kV Tap	Eaton County	MI	NRIS
J1327	Battery Storage	20	METC	Marquette - Vergennes 138 kV Line Tap	Ionia County	MI	NRIS
J1329	Battery Storage	20	ITC	Majestic - Milan 345 kV Line Tap	Washtenaw County	MI	NRIS
J1331	Solar	100	ITC	Greenwood 120 kV Substation	St. Clair County	MI	NRIS
J1350	Solar	150	METC	Milan - Monroe - Morocco 345 kV Point	Monroe County	MI	NRIS
J1375	Solar	100	METC	Du Pont - Cobb - White Lake 138kv	Muskegon County	MI	NRIS
J1379	Solar	200	METC	Eureka - Vestaburg 138 kV Line Tap	Montcalm County	MI	NRIS
J1389	Solar	120	METC	Murphy Station - Nelson Road 345 kV Line tap	Saginaw County	MI	NRIS
J1394	Solar	150	METC	Cornell - Layton 138kV Line	Shiawassee County	MI	NRIS
J1399	Solar	90	METC	Delhi - Churchill - Tompkins 138kV Line	Ingham County	MI	NRIS
J1401	Solar	150	METC	Batavia - Wagner 138kV Line	Branch County	MI	NRIS
J1403	Solar	200	METC	Marquette 138 kV Substation	Ionia County	MI	NRIS
J1430	Solar	100	METC	Verona - Foundary 138kV Line	Calhoun County	MI	NRIS
J1449	Solar	100	METC	Morocco 138kV Substation	Monroe County	MI	NRIS
J1472	Solar	100	METC	Leoni - Parr Road 138kV line	Jackson County	MI	NRIS

1.2. Total Network Upgrades

The cost allocation of Network Upgrades for the projects in the DPP 2019 Cycle1 Michigan Phase 1 reflects responsibilities for mitigating system impacts. The total cost of network upgrades is listed in Table 2 below. The costs for Network Upgrades are planning-level estimates and subject to revision in the facility studies. Details pertaining to the cost allocation has been detailed in [Appendix A – Cost Allocation Summary \(CEII\)](#)

Table 2: Total Cost of Network Upgrades for DPP 2019 Cycle1 Michigan Phase 1 Projects

Project	DPP Phase I									Payment in DPP Phase I		Payment in DPP Phase II		Interconnection Facilities - Self Fund
	ERIS Network Upgrades (\$)				NRIS Network Upgrades (\$)	Interconnection Facilities (\$)		Shared Network Upgrades (\$)	Total Cost of NU (\$)	M2 (\$)	M3 =10% of NU-M2 (\$)	20% of Total Cost of NU (\$)	M4= 20% of NU-M2-M3 (\$)	
	Thermal	Stability	Short Circuit	Affected System	Deliverability	TO Network Upgrades	TO-Owned Direct Assigned							
a	b	c	d	e	f	g	h	i	j = b+c+d+f+g+i					
J1192						\$0.00	\$0.00		\$0.00	\$140,000	\$0.00			NA
J1194						\$2,000,250.00	\$299,214.00		\$2,000,250.00	\$900,800	\$0.00			No
J1196						\$10,300,000.00	\$500,000.00		\$10,300,000.00	\$800,000	\$230,000.00			No
J1203						\$4,900,000.00	\$500,000.00		\$4,900,000.00	\$500,000	\$0.00			No
J1206					\$280,257.14	\$350,000.00	\$250,000.00		\$630,257.14	\$200,000	\$0.00			No
J1207					\$280,257.14	\$350,000.00	\$250,000.00		\$630,257.14	\$200,000	\$0.00			No
J1210						\$6,800,000.00	\$500,000.00		\$6,800,000.00	\$200,000	\$480,000.00			No
J1211					\$1,000,000.00	\$16,400,000.00	\$500,000.00		\$17,400,000.00	\$200,000	\$1,540,000.00			No
J1224						\$5,150,000.00	\$250,000.00		\$5,150,000.00	\$680,000	\$0.00			No
J1226						\$5,600,000.00	\$500,000.00		\$5,600,000.00	\$800,000	\$0.00			No
J1248	\$2,007,799.74					\$5,100,000.00	\$500,000.00		\$7,107,799.74	\$400,000	\$310,779.97			No
J1255						\$10,016,666.67	\$250,000.00		\$10,016,666.67	\$800,000	\$201,666.67			No
J1262						\$0.00	\$0.00		\$0.00	\$100,000	\$0.00			NA
J1292						\$11,566,666.67	\$500,000.00		\$11,566,666.67	\$789,600	\$367,066.67			No
J1297					\$4,900,000.00	\$0.00	\$0.00		\$4,900,000.00	\$680,000	\$0.00			NA
J1307						\$10,300,000.00	\$500,000.00		\$10,300,000.00	\$600,000	\$430,000.00			No
J1310	\$12,504,112.95					\$5,100,000.00	\$500,000.00		\$17,604,112.95	\$500,000	\$1,260,411.30			No
J1319						\$0.00	\$0.00		\$0.00	\$487,200	\$0.00			NA
J1320	\$1,741,771.43					\$5,100,000.00	\$500,000.00		\$6,841,771.43	\$600,000	\$84,177.14			No
J1322						\$10,300,000.00	\$500,000.00		\$10,300,000.00	\$800,000	\$230,000.00			No
J1327						\$10,016,666.67	\$250,000.00		\$10,016,666.67	\$80,000	\$921,666.67			No
J1329						\$5,150,000.00	\$250,000.00		\$5,150,000.00	\$80,000	\$435,000.00			No
J1331						\$4,200,000.00	\$500,000.00		\$4,200,000.00	\$400,000	\$20,000.00			No
J1350						\$10,300,000.00	\$500,000.00		\$10,300,000.00	\$600,000	\$430,000.00			No
J1375						\$7,500,000.00	\$500,000.00		\$7,500,000.00	\$400,000	\$350,000.00			No
J1379					\$10,000,000.00	\$5,100,000.00	\$500,000.00		\$15,100,000.00	\$800,000	\$710,000.00			No
J1389						\$10,300,000.00	\$500,000.00		\$10,300,000.00	\$480,000	\$550,000.00			No
J1394	\$52,300,000.00					\$0.00	\$0.00		\$52,300,000.00	\$600,000	\$4,630,000.00			NA
J1399						\$0.00	\$0.00		\$0.00	\$360,000	\$0.00			NA
J1401					\$1,077,800.00	\$5,100,000.00	\$500,000.00		\$6,177,800.00	\$600,000	\$17,780.00			No
J1403						\$500,000.00	\$500,000.00		\$500,000.00	\$800,000	\$0.00			No
J1430	\$2,259,803.49				\$200,000.00	\$1,000,000.00	\$500,000.00		\$3,459,803.49	\$400,000	\$0.00			No
J1449						\$500,000.00	\$500,000.00		\$500,000.00	\$400,000	\$0.00			No
J1472	\$10,973,597.36				\$354,686.47	\$5,100,000.00	\$500,000.00		\$16,428,283.82	\$400,000	\$1,242,828.38			No



Analyses performed demonstrate the following transmission facilities are required to reliably interconnect this group of generators to the transmission system. Energy Resource Interconnection Service (ERIS) Network Upgrades and Network Resource Interconnection Service (NRIS) Network Upgrades are shown in Table 3. No Shared Network Upgrades was identified for this cycle.

Table 3: ERIS & NRIS Upgrades (Planning level cost estimates)

Element Name	Network Upgrade	TO	GI projects requiring upgrade for ERIS	GI projects requiring upgrade for NRIS	Cost of solution (\$)	MTEP project ISD	Self - Refund
Wagner - Alder Creek Junction 138kV	MTEP 17 Project 11884 in service 5/14/2021 upgrades conductor to 954 ACSR. METC will need to upgrade station equipment to raise limit to 288 MVA SN, 332 MVA SE	METC		J1430	\$100,000		No
Summerton - Bluegrass Junction 138kV	Reconductor 11.1 miles of 336 ACSR with 795ACSS. (416MVA SN, 416 MVA SE)	METC		J1379	\$5,000,000		No
Claremont - Layton 138kV	Rebuild about 17.1 miles of 477 ACSR to 954 ACSR. (288 MVA SN, 332 MVA SE)	METC	J1394		\$27,500,000		No
Eureka - Derby Junction 138kV	MTEP 21 Target A Project 20016 Rebuilds the 336 to 954 ACSR. (288 MVA SN, 332 MVA SE)	METC	J1379	J1379,J1211	\$0	6/1/2025	N/A
Goss - Pasadena Junction 138kV	Raise the sag limit of 2.2 miles of 795 ACSR to 280 MVA	METC		J1297,J1394	\$0		N/A
Moore Road - Ash Road Junction 138kV	MTEP 20 project 18084 Raises sag to 166 MVA	METC	J1320		\$0		N/A
Leoni - Washington Junction 138kV	Rebuild about 13 miles of 477 ACSR to 954 ACSR. (288 MVA SN, 331 MVA SE)	METC	J1472,J1310	J1472,J1310	\$21,000,000		No
Cornell - North Corunna Junction 138kV	Above Conductor Max. Rebuild 3 miles of 477 to 954 ACSR. (288 MVA SN, 332 MVA SE)	METC		J1297	\$4,900,000		No
Alder Creek - Barnum Junction 138kV	MTEP 17 Project 11884 in service 5/14/2021 upgrades conductor to 954 ACSR. METC will need to upgrade station equipment to raise limit to 288 MVA SN, 332 MVA SE	METC		J1430	\$100,000		No
Lark - Washington Junction 138kV	Raise sag limit to 195 MVA	METC/ITC	J1472,J1310	J1472,J1310,J1399	\$0		N/A
Ash Rd Jn to Moore Rd 138 kV	MTEP 20 project 18084 Raises sag to 166 MVA	METC		J1320,J1401,J1206,J1207	\$0	12/31/2023	N/A
Coldwater - Ash Road Junction 138kV		METC	J1320,J1401,J1206,J1207		\$0	12/31/2023	N/A
Wagner to J1401 138 kV		METC		J1206,J1207,J1430	\$0	5/14/2021	N/A
Cold Water to Ash Rd Jn 138 kV	MTEP 17 Project 11884 in service 5/14/2021 raises limit to 288 MVA SN, 332 MVA SE	METC		J1248	\$0	5/14/2021	N/A
Luce - Regal 138kV	Reconductor 0.75 miles of 336 ACSR with 795 ACSS. (288 MVA SN, 332 MVA SE)	METC		J1211	\$1,000,000		No
Luce - Bluegrass Junction 138kV	Reconductor 11.1 miles of 336 ACSR with 795ACSS. (416MVA SN, 416 MVA SE)	METC		J1379	\$5,000,000		No
Benston Junction - Peterson Junction 138kV	Raise the sag limit of the 336 ACSR to 111 MVA	METC		J1375	\$0		N/A
Cobb - Peterson Junction 138kV	Raise the sag limit of the 336 ACSR to 111 MVA	METC		J1375	\$0		N/A
Cold Water to Ash Rd Jn 138 kV	MTEP 20 project 18084 Raises sag to 166 MVA	METC		J1320,J1401,J1206,J1207	\$0	12/31/2023	N/A
Cement City Junction - Seamless 138kV	MTEP 20 15939 increases the limit to 227 MVA SN, 286 MVA SE	METC		J1472,J1310,J1399,J1248	\$0	12/31/2022	N/A
Monroe - Lalendorf 345kV	MTEP 20116 Raises limits to 1486 MVA SN, 1824 MVA SE	CE/ITC		J1226,J1196,J1194,J1224,J1350,J1297,J1394,J1322,J1319,J1379,J1208,J1331,J1449,J1403,J1389,J1255,J1307,J1292,J1472,J1310,J1320,J1399,J1401,J1248,J1430,J1375,J1210,J1211,J1192,J1206,J1207,J1329,J1327	\$0	12/31/2024	N/A
Caniff 120kV phase shifter	Can be fixed by operating phase shifter	ITC		J1196,J1319,J1331,J1203	\$0		N/A
Caniff 345 / 120kV transformer	Can be fixed by operating phase shifter	ITC		J1196,J1319,J1331,J1203	\$0		N/A
J1088 - Layton 138kV	Rebuild about 6.9 miles of 477 ACSR to 954ACSR. (288 MVA SN, 332 MVA SE)	METC	J1394	J1394	\$11,100,000		No
Cornell - J1088 138kV	Rebuild about 8.5 miles of 477 ACSR to 954ACSR. (288 MVA SN, 332 MVA SE)	METC	J1394	J1394	\$13,700,000		No
J1207 - Newton 138kV	J1207 Gen Tie	METC	J1206,J1207		\$0		N/A
Coldwater - J1320 138kV	Rebuild 2 miles of 336 and 477 ACSR to 954 ACSR. (288 MVA SN, 332 MVA SE)	METC	J1320	J1320,J1401,J1206,J1207	\$3,400,000		No
J1379 - Deja Junction 138kV	MTEP 21 Target A Project 20016 Rebuilds the 336 to 954 ACSR. (288 MVA SN, 332 MVA SE)	METC	J1379,J1403,J1403	J1379,J1403,J1255,J1327	\$0 if approved/ \$18,700,000.00 (if not approved)	6/1/2025	N/A
J1379 - Derby Junction 138kV		METC	J1379	J1379,J1211			No
Batavia - J1401 138kV	MTEP 17 Project 11884 in service 5/14/2021 raises limit to 288 MVA SN, 332 MVA SE	METC	J1401,J1206,J1207	J1401,J1206,J1207,J1480,J1248	\$0	5/14/2021	N/A
J1401 - Wagner 138kV		METC	J1401,J1320	J1401,J1320	\$0	5/14/2021	N/A
J1472 - Leoni 138kV	Raise the sag limits of the 795 ACSR and the 636 ACSR to 194 MVA	METC		J1472	\$0		N/A
J758 (Mud Lake) - Verona #2 138kV	Rebuild about 4.4 miles of 336 ACSR to 954 ACSR. (288 MVA SN, 332 MVA SE)	METC	J1310,J1430,J1248	J1310,J1430,J1248,J1472	\$7,100,000		No

Note:

- 1) Details pertaining to upgrades, costs, and the execution plan for interconnection of the generating facility at the POI will be documented in the Facility Study for Interconnecting Generator.
- 2) Facilities that have been included as base case assumptions and the level of interconnection service that would be conditional upon these facilities being in service will be documented in the GIA (Generator Interconnection Agreement) for each respective GI request successfully achieving GIA execution.
- 3)
- 4) Analysis performed shows there are no study projects for Shared Network Upgrade cost allocation.
- 5) A detailed summary of the estimated cost of Network Upgrades on a per project basis can be found in Appendix B – Network Upgrade Table Per Project (CEII).

2. FERC Order 827 Compliance Review

The Final Rule of FERC Order 827 “Reactive Power Requirements for Non-Synchronous Generation”, which was issued June 16, 2016, stated that “Under this Final Rule, newly interconnecting non-synchronous generators that have not yet executed a Facilities Study Agreement as of the effective date of this Final Rule will be required to provide dynamic reactive power within the range of 0.95 leading to 0.95 lagging at the high-side of the generator substation.” As such, this Final Rule applies to J1192, J1194, J1196, J1203, J1206, J1207, J1210, J1211, J1224, J1226, J1248, J1255, J1262, J1292, J1297, J1307, J1310, J1319, J1320, J1322, J1327, J1329, J1331, J1350, J1370, J1375, J1379, J1389, J1394, J1399, J1401, J1403, J1430, J1449 and J1472 projects included in the DPP 2019 Cycle1 Michigan study cycle.

In this study, the power factor at the high-side of the generator substation for each inverter based project was calculated and reviewed. The study method is to set Qgen of each study project at its Qmax, solve the case, then record the P and Q injection on the high side of the generator substation to calculate the lagging power factor (injecting VAR to the system). The same process is then repeated by setting Qgen at Qmin to calculate the leading power factor (absorbing VAR from the system).

The results show that all projects meet the requirement to maintain 0.95 leading power factor, however, four projects do not meet the requirement to provide reactive power capability corresponding to 0.95 lagging power factor, as highlighted in red below in Table 4. Additional reactive support will be needed for these projects to meet the FERC requirement on reactive power capability prior to the completion of their GIA.

Table 4: FERC Order 827 Review Results

Project	Pmax (MW)	Reactive Power Capability (MVar)	Proposed VAR Compensation	VAR Injection			VAR Absorption			Meet FERC Order 827 Requirement?	Add'l VAR Needed (MVar)
				P (MW)	Q (MVar)	Lagging p.f (pu)	P (MW)	Q (MVar)	Leading p.f (pu)		
J1192	35	+/-11.504	1 X 7	34.6	17.56	0.8917308	34.42	-14.96	-0.9171209	Yes	-
J1194	225.2	+/-109.07	#N/A	221.05	80.3	0.939905	220.28	-77.31	-0.9435749	Yes	-
J1196	200	+/-108.00	#N/A	200.25	70.7	0.94	199.57	-179.2	-0.74	Yes	-
J1203	125	+/-45.5	4 X 12	122.73	109.37	0.746573	123.17	-44.29	-0.941012	Yes	-
J1206	50	+/-37.495	#N/A	50.07	45.5	0.7400738	50.07	-18.75	-0.9364905	Yes	-
J1207	50	+/-37.495	#N/A	50.07	37.49	0.8004794	50.07	-18.75	-0.9364905	Yes	-
J1210	50	+/-87.815	1 X 10	50.19	48.58	0.7185375	49.33	-52.37	-0.6856638	Yes	-
J1211	50	+/-87.815	1 X 10	49.65	99.49	0.44653	49.23	-52.05	-0.6871521	Yes	-
J1224	170	+/-82.335	1 X 5	187.22	41.24	0.976588	187.3	-146.71	-0.7872451	No	20.3
J1226	200	+/-150	2 X 13	98.48	123.44	0.6236442	98.36	-40.58	-0.9244171	Yes	-
J1248	100	+/-78.75	#N/A	100.38	71.43	0.8147687	100.11	-46	-0.9086647	Yes	-
J1255	200	+/-65.736	1 X 3	99.34	80.78	0.7758607	99.29	-38.89	-0.9311237	Yes	-
J1262	25	+/-10	#N/A	398.85	135.82	0.94662	399.32	133.8	-0.9481882	Yes	-
J1292	198	+/-13.2911	#N/A	196.5	70.19	0.941725	194.53	-116.08	-0.8587329	Yes	-
J1297	170	+/-96.343	#N/A	145.34	26	0.9843731	-142.33	116	0.7751619	No	21.8
J1307	150	+/-85.009	#N/A	148.09	84.27	0.8691338	147.51	-58.32	-0.9299562	Yes	-
J1310	125	+/-41.09	1 X 42	123.46	135.41	0.6737477	122.99	-51.57	-0.9222116	Yes	-
J1319	117.5	+/-61.045	1 X 2	118.45	46.2	0.931643	115	-69	-0.8574929	Yes	-

Project	Pmax (MW)	Reactive Power Capability (MVar)	Proposed VAR Compensation	VAR Injection			VAR Absorption			Meet FERC Order 827 Requirement?	Add'l VAR Needed (MVar)
				P (MW)	Q (MVar)	Lagging p.f (pu)	P (MW)	Q (MVar)	Leading p.f. (pu)		
J1320	150	+/-61.616	#N/A	148.58	10.25	0.9976289	147.16	-73.67	-0.8942083	No	38.6
J1322	200	+/-82.155	#N/A	197.96	59.1	0.9582091	196.25	-96.67	-0.8970717	No	6
J1327	20	+/-9.686	#N/A	19.75	30.95	0.5379325	19.84	-10.96	-0.8753202	Yes	-
J1329	20	+/-9.686	#N/A	19.11	75.03	0.2468182	19.83	-11	-0.8744691	Yes	-
J1331	100	+/-32.87	1 X 28	99.5	40.97	0.92468	99.34	-36.73	-0.9379411	Yes	-
J1350	150	+/-61.616	1 X 28	148.61	63.66	0.9192122	147.78	-69.48	-0.9049685	Yes	-
J1375	100	+/-75	2 X 10	98.93	93.69	0.7260742	98.65	-38.42	-0.9318255	Yes	-
J1379	200	+/-94.014	#N/A	197.65	64.88	0.9501203	195.77	-112.72	-0.8666149	No	0.1
J1389	120	+/-61.539	#N/A	118.18	90.39	0.7943025	117.8	-69.56	-0.8610837	Yes	-
J1394	150	+/-85.009	#N/A	148.31	52.13	0.9434182	147.67	-56.38	-0.9342249	Yes	-
J1399	90	+/-51.005	#N/A	88.64	76.95	0.7551463	88.69	-33.56	-0.9352806	Yes	-
J1401	150	+/-85.009	#N/A	148.39	42.08	0.9620652	147.27	-59.06	-0.928146	No	6.7
J1403	200	+/-110.426	#N/A	98.68	79.59	0.778377	98.71	-32.71	-0.9492396	Yes	-
J1430	100	+/-32.868	2 X 7	99.18	66.62	0.8301138	98.95	-39.4	-0.9290583	Yes	-
J1449	100	+/-32.868	2 X 7	98.9	32.38	0.950361	98.75	-34.52	-0.943985	No	67
J1472	100	+/-32.868	2 X 7	48.32	25.07	0.8876405	48.74	-25.94	-0.8827637	Yes	-

3. Model Development and Study Assumptions

3.1. Base Case Models

The origin of the DPP 2019 Cycle1 Michigan models is the MTEP 19 models with the Bench Cases including all pre-queued projects and associated network upgrades, while the Study Cases contain all of the interconnection requests in DPP 2019 Cycle1 Michigan Phase 1, in addition to all the facilities in the Bench Cases. Table 5 provides the list of different Scenarios study for DPP 2019 Cycle1 Michigan.

- Bench Cases
 - DPP19C1-Phasel-2024SH-Bench
 - DPP19C1-Phasel-2024SUM-Bench
- Study Cases
 - DPP19C1-Phasel-2024SH-Study
 - DPP19C1-Phasel-2024SUM-Study

Table 5: DPP 2019 Cycle1 Michigan-Scenarios

Case Category	Case Name	Case Season	Flow on ITCT-IESO interface	Flow on METC-ATC DC line	Status of Ludington	Load Level
1	DPP19C1-Phasel-2024SUM-Study	Summer Peak 2024	Zero Flow	5 MW ATC to METC	Generating (6 Gen)	Peak
2	DPP19C1-Phasel-2024SH-Study	Summer Shoulder 2024	Zero Flow	5 MW ATC to METC	Generating (5 Gen)	MTEP 18 Base SH (77-78%)

3.2. Monitored Elements

Under NERC category P0 conditions (system intact) branches were monitored for loading above the normal rating (PSS®E Rating A), and for NERC category P1-P7 conditions branches were monitored for emergency rating (PSS®E Rating B). Voltage limits were specified for system intact and contingent conditions as per applicable Transmission Owner Planning Criteria.

3.3. Contingencies

The following contingencies were considered in the steady state analysis:

- 1) NERC Category P0 (system intact -- no contingencies)
- 2) NERC Category P1 contingencies
 - a. Single element outages, at buses with a nominal voltage of 68 kV and above
 - b. Multiple element NERC Category P1 contingencies
- 3) NERC Category P2-P7 contingencies
- 4) For all the contingencies and post-disturbance analyses, cases were solved with transformer tap adjustment enabled, area interchange adjustment disabled, phase shifter adjustment disabled (fixed) and switched shunt adjustment enabled.

3.4. Study Methodology

Non-linear (AC) contingency analysis was performed on the benchmark and study cases, and the incremental impact of the DPP 2019 Cycle1 Michigan generating facilities was evaluated by comparing the steady state performance of the transmission system in the Bench and Study Cases. Analyses used PSS®E version 34.6.1 and TARA version 2001_2_0_0.

3.5. Performance Criteria

A branch is considered a thermal constraint if the following conditions are met:

- 1) The generator has a larger than twenty percent (20%) sensitivity factor on the overloaded facilities under post-contingent condition (see NERC TPL) or five percent (5%) sensitivity factor under system-intact condition, or
- 2) The overloaded facility or the overload-causing contingency is at generator's outlet, or
- 3) The megawatt impact due to the generator is greater than or equal to twenty percent (20%) of the applicable rating (normal or emergency) of the overloaded facility, or
- 4) For any other constrained facility, where none of the Study Generators meet one of the above criteria, however, the cumulative MW impact of the group of study generators is greater than twenty percent (20%) of the rating of the facility, then only those study generators whose individual MW impact is greater than five percent (5%) of the rating of the facility and has DF greater than five percent (5%) will be responsible for mitigating the cumulative MW impact constraint, or
- 5) Impacts on Affected Systems would be classified as Injection constraints based on the Affected Systems' criteria, or
- 6) Any other applicable Transmission Owner FERC filed Local Planning Criteria are met.

A bus is considered a voltage constraint if both of the following conditions are met:

- 1) The bus voltage is outside of the applicable normal or emergency limits for the post change case, and
- 2) The change in bus voltage is greater than 0.01 per unit

All generators must mitigate thermal injection constraints and voltage constraints in order to obtain any type of Interconnection Service. Further, all generators requesting Network Resource Interconnection Service (NRIS) must mitigate constraints found by using the Deliverability algorithm, to meet the system performance criteria for NERC category P1 events, if DFAX due to the study generator is equal to or greater than 5%.

4. Thermal Analysis

The thermal analysis results for 2024 Summer Peak and Summer Shoulder scenarios show generator projects J1248, J1310, J1320, J1394, J1430 and J1472 causing constraints. The details pertaining to the



thermal analysis can be found in Appendix A – Cost Allocation Summary (CEII)

4.1. J1248

The thermal analysis identified the generator to contribute to the following constraint:

1. J758 (Mud Lake) - Verona #2 138kV

Per MISO cost allocation rules, the project receives cost allocation for upgrades required to mitigate the above listed constraint. The planned upgrade is to rebuild the line. The total cost estimate is \$7.1 million, with the project allocated 28.28% of the cost resulting \$2,007,799.74 million.

4.2. J1310

The thermal analysis identified the generator to contribute to the following constraint:

1. J758 (Mud Lake) - Verona #2 138kV

Per MISO cost allocation rules, the project receives cost allocation for upgrades required to mitigate the above listed constraint. The planned upgrade is to rebuild the line. The total cost estimate is \$7.1 million, with the project allocated 34.90% of the cost resulting \$2,477,710.31 million.

2. Leoni to Washington Jn 138kV

Per MISO cost allocation rules, the project receives cost allocation for upgrades required to mitigate the above listed constraint. The planned upgrade is to rebuild the line. The total cost estimate is \$21 million, with the project allocated 47.7% of the cost resulting \$10,026,402.64 million.

4.3. J1320

The thermal analysis identified the generator to contribute to the following constraint:

1. J1320 substation to Coldwater 138kV

Per MISO cost allocation rules, the project receives cost allocation for upgrades required to mitigate the above listed constraint. The planned upgrade is to rebuild the line. The total cost estimate is \$3.4 million, with the project allocated 51% of the cost resulting \$1,741,771.43 million.

4.4. J1394

The thermal analysis identified the generator to contribute to the following constraint:

1. J1088 POI substation to Layton 138kV

Per MISO cost allocation rules, the project receives cost allocation for upgrades required to mitigate the above listed constraint. The planned upgrade is to rebuild the line. The total cost estimate is \$11.1 million, with the project allocated 100% of the cost resulting \$11.1 million.

2. Cornell - J1088 138kV

Per MISO cost allocation rules, the project receives cost allocation for upgrades required to mitigate the above listed constraint. The planned upgrade is to rebuild the line. The total cost estimate is \$13.7 million, with the project allocated 100% of the cost resulting \$13.7 million.

3. Claremont to Layton 138kV



Per MISO cost allocation rules, the project receives cost allocation for upgrades required to mitigate the above listed constraint. The planned upgrade is to rebuild the line. The total cost estimate is \$27.5 million, with the project allocated 100% of the cost resulting \$27.5 million.

4.5. J1430

1. J758 (Mud Lake) - Verona #2 138kV

Per MISO cost allocation rules, the project receives cost allocation for upgrades required to mitigate the above listed constraint. The planned upgrade is to rebuild the line. The total cost estimate is \$7.1 million, with the project allocated 31.83% of the cost resulting \$2,259,803.49 million.

4.6. J1472

1. Leoni to Washington Jn 138kV

Per MISO cost allocation rules, the project receives cost allocation for upgrades required to mitigate the above listed constraint. The planned upgrade is to rebuild the line. The total cost estimate is \$21 million, with the project allocated 52.26% of the cost resulting \$10,973,597.36 million.

5. Voltage Analysis

The voltage analysis results for 2024 Summer Peak and Summer Shoulder scenarios show that the study generators do not cause any voltage violations. The details pertaining to the voltage analysis can be found in Appendix C – MISO ERIIS Analysis (CEII).

6. Stability Analysis

Stability analysis will be performed in phase 2 of the study.

7. Short Circuit Analysis

Short Circuit analysis will be performed in phase 2 of the study.

8. Affected System Impact Study

PJM, CE, DTE, and WPSC Affected Systems studies will be performed in phase 2 of the study.

9. Deliverability Analysis

9.1. Introduction

Generator interconnection projects have to pass Generator Deliverability Study to be granted NRIS. If the generator is deemed not fully deliverable, the customer can choose either to change the project to an Energy Resource (ER) project or to proceed with the system upgrades that will make the generator fully deliverable. Generator Deliverability Study ensures that the Network Resources, on an aggregate basis, can meet the MISO aggregate load requirements during system peak condition without getting bottlenecked. The wind generators are tested at 100% of their maximum output level which then can be used to meet Resource Adequacy obligations, under Module E, of the MISO Transmission and Energy Market Tariff (TEMT).

MISO Generator Deliverability Study whitepaper describing the algorithm can be found in BPM 015 –



Generation Interconnection, Appendix C.

9.2. Determining the MW Restriction

If one facility is overloaded based on the assessed “severe yet credible dispatch” scenario described in the study methodology, and the generator under study has a DF greater than 5%, part or all of its output is not deliverable. The restricted MW is calculated as following:

$$(\text{MW restricted}) = (\text{worst loading} - \text{MW rating}) / (\text{generator sensitivity factor})$$

If the result is larger than the maximum output of the generator, 100% of this generator’s output is not deliverable.

9.3. Deliverability Study Results

The limiting constraints (mon-con pairs) seen in the deliverability analysis for the 2024 Summer case are summarized in Appendix J – MISO Deliverability Analysis (CEII)

9.4. J1192

This generator is determined to be fully deliverable for 35 MW, contingent upon the system upgrades and contingent facilities identified in the NRIS analysis. Table 6: NRIS Results for J1192 shows the NRIS results and cost estimates determined in the NRIS analysis.

Table 6: NRIS Results for J1192

J1192 Deliverable (NRIS) Amount in 2024 Case: (Conditional on ERIIS and case assumptions)						29.98 MW (85.66%)			
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Mitigation /Upgrade	MTEP project ISD	Level of Service Attainable (MW)	Distribution Factor	Constraint in ERIIS Analysis?	Projects Associated with ERIS Constraint	Projects Associated with NRIS Constraint	Upgrade Cost Allocated to Project	Total Cost of Upgrade (\$)
Monroe - Lalendorf 345kV	MTEP 20116 Raises limits to 1486 MVA SN, 1824 MVA SE	12/31/2024	35	0.1772			J1226, J1196, J1194, J1224, J1350, J1297, J1394, J1322, J1319, J1379, J1203, J1331, J1449, J1403, J1389, J1255, J1307, J1292, J1472, J1310, J1320, J1399, J1401, J1248, J1430, J1375, J1210, J1211, J1192, J1206, J1207, J1329, J1327	\$0.00	\$0.00

9.5. J1194

This generator is determined to be fully deliverable for 225.2 MW, contingent upon the system upgrades and contingent facilities identified in the NRIS analysis. Table 7 shows the NRIS results and cost estimates determined in the NRIS analysis.

Table 7: NRIS Results for J1194

J1194 Deliverable (NRIS) Amount in 2024 Case: (Conditional on ERIIS and case assumptions)						192.92 MW (85.67%)			
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Mitigation /Upgrade	MTEP project ISD	Level of Service Attainable (MW)	Distribution Factor	Constraint in ERIIS Analysis?	Projects Associated with ERIS Constraint	Projects Associated with NRIS Constraint	Upgrade Cost Allocated to Project	Total Cost of Upgrade (\$)

Monroe - Lalendorf 345kV	MTEP 20116 Raises limits to 1486 MVA SN, 1824 MVA SE	12/31/2024	225.2	0.2088		J1226, J1196, J1194, J1224, J1350, J1297, J1394, J1322, J1319, J1379, J1203, J1331, J1449, J1403, J1389, J1255, J1307, J1292, J1472, J1310, J1320, J1399, J1401, J1248, J1430, J1375, J1210, J1211, J1192, J1206, J1207, J1329, J1327	\$0.00	\$0.00
--------------------------	--	------------	-------	--------	--	---	--------	--------

9.6. J1196

This generator is determined to be fully deliverable for 200 MW, contingent upon the system upgrades and contingent facilities identified in the NRIS analysis. Table 8 shows the NRIS results and cost estimates determined in the NRIS analysis.

Table 8: NRIS Results for J1196

J1196 Deliverable (NRIS) Amount in 2024 Case: (Conditional on ERIIS and case assumptions)						83.49 MW (41.74%)			
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Mitigation / Upgrade	MTEP project ISD	Level of Service Attainable (MW)	Distribution Factor	Constraint in ERIIS Analysis?	Projects Associated with ERIS Constraint	Projects Associated with NRIS Constraint	Upgrade Cost Allocated to Project	Total Cost of Upgrade (\$)
Caniff 120kV phase shifter	Can be fixed by operating phase shifter		110.45	0.065			J1196, J1319, J1331, J1203	\$0.00	\$0.00
Caniff 345 / 120kV transformer	Can be fixed by operating phase shifter		171.33	0.065			J1196, J1319, J1331, J1203	\$0.00	\$0.00
Monroe - Lalendorf 345kV	MTEP 20116 Raises limits to 1486 MVA SN, 1824 MVA SE	12/31/2024	200	0.2635			J1226, J1196, J1194, J1224, J1350, J1297, J1394, J1322, J1319, J1379, J1203, J1331, J1449, J1403, J1389, J1255, J1307, J1292, J1472, J1310, J1320, J1399, J1401, J1248, J1430, J1375, J1210, J1211, J1192, J1206, J1207, J1329, J1327	\$0.00	\$0.00

9.7. J1203

This generator is determined to be fully deliverable for 125 MW, contingent upon the system upgrades and contingent facilities identified in the NRIS analysis. Table 9 shows the NRIS results and cost estimates determined in the NRIS analysis.

Table 9: NRIS Results for J1203

J1203 Deliverable (NRIS) Amount in 2024 Case: (Conditional on ERIIS and case assumptions)						52.18 MW (41.74%)			
Next Upgrade for Higher NRIS Level									

(cumulative) (i.e. All upgrades must be made for 100% NRIS)	Mitigation /Upgrade	MTEP project ISD	Level of Service Attainable (MW)	Distribution Factor	Constraint in ERS Analysis?	Projects Associated with ERIS Constraint	Projects Associated with NRIS Constraint	Upgrade Cost Allocated to Project	Total Cost of Upgrade (\$)
Caniff 120kV phase shifter	Can be fixed by operating phase shifter		69.03	0.0521			J1196, J1319, J1331, J1203	\$0.00	\$0.00
Caniff 345 / 120kV transformer	Can be fixed by operating phase shifter		107.08	0.0521			J1196, J1319, J1331, J1203	\$0.00	\$0.00
Monroe - Lalendorf 345kV	MTEP 20116 Raises limits to 1486 MVA SN, 1824 MVA SE	12/31/2024	125	0.2308			J1226, J1196, J1194, J1224, J1350, J1297, J1394, J1322, J1319, J1379, J1203, J1331, J1449, J1403, J1389, J1255, J1307, J1292, J1472, J1310, J1320, J1399, J1401, J1248, J1430, J1375, J1210, J1211, J1192, J1206, J1207, J1329, J1327	\$0.00	\$0.00

9.8. J1206

This generator is determined to be fully deliverable for 50 MW, contingent upon the system upgrades and contingent facilities identified in the NRIS analysis. Table 10 shows the NRIS results and cost estimates determined in the NRIS analysis.

Table 10: NRIS Results for J1206

J1206 Deliverable (NRIS) Amount in 2024 Case: (Conditional on ERS and case assumptions)						14.99 MW (29.98%)			
Next Upgrade for Higher NRIS Level									
(cumulative) (i.e. All upgrades must be made for 100% NRIS)	Mitigation /Upgrade	MTEP project ISD	Level of Service Attainable (MW)	Distribution Factor	Constraint in ERS Analysis?	Projects Associated with ERIS Constraint	Projects Associated with NRIS Constraint	Upgrade Cost Allocated to Project	Total Cost of Upgrade (\$)
Batavia - J1401 138kV	MTEP 17 Project 11884 in service 5/14/2021 raises limit to 288 MVA SN, 332 MVA SE	5/14/2021	32.93	0.4534	yes	J1401, J1206, J1207	J1401, J1206, J1207, J1430, J1248	\$0.00	\$0.00
Wagner to J1401 138 kV	MTEP 17 Project 11884 in service 5/14/2021 raises limit to 288 MVA SN, 332 MVA SE	5/14/2021	42.83	0.4534			J1206, J1207, J1430	\$0.00	\$0.00
Monroe - Lalendorf 345kV	MTEP 20116 Raises limits to 1486 MVA SN, 1824 MVA SE	12/31/2024	44.54	0.1219			J1226, J1196, J1194, J1224, J1350, J1297, J1394, J1322, J1319, J1379, J1203, J1331, J1449, J1403, J1389, J1255, J1307, J1292, J1472, J1310, J1320, J1399, J1401, J1248, J1430, J1375, J1210, J1211, J1192, J1206, J1207, J1329, J1327	\$0.00	\$0.00
Cold Water to Ash Rd Jn 138 kV	MTEP 20 project 18084 Raises sag to 166 MVA	12/31/2023	47.21	0.1912			J1320, J1401, J1206, J1207	\$0.00	\$0.00
Ash Rd Jn to Moore Rd 138 kV	MTEP 20 project 18084 Raises sag to 166 MVA	12/31/2023	49.76	0.1912			J1320, J1401, J1206, J1207	\$0.00	\$0.00

Coldwater - J1320 138kV	Reuild 2 miles of 336 and 477 ACSR to 954 ACSR. (288 MVA SN, 332 MVA SE)		50	0.1155	yes	J1320	J1320, J1401, J1206, J1207	\$280,257.14	\$3,400,000.00
-------------------------	--	--	----	--------	-----	-------	----------------------------	--------------	----------------

9.9. J1207

This generator is determined to be fully deliverable for 50 MW, contingent upon the system upgrades and contingent facilities identified in the NRIS analysis. Table 11 shows the NRIS results and cost estimates determined in the NRIS analysis.

Table 11: NRIS Results for J1207

J1207 Deliverable (NRIS) Amount in 2024 Case: (Conditional on ERIS and case assumptions)						14.99 MW (29.98%)			
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Mitigation /Upgrade	MTEP project ISD	Level of Service Attainable (MW)	Distribution Factor	Constraint in ERIS Analysis?	Projects Associated with ERIS Constraint	Projects Associated with NRIS Constraint	Upgrade Cost Allocated to Project	Total Cost of Upgrade (\$)
Batavia - J1401 138kV	MTEP 17 Project 11884 in service 5/14/2021 raises limit to 288 MVA SN, 332 MVA SE	5/14/2021	32.93	0.4534	yes	J1401, J1206, J1207	J1401, J1206, J1207, J1430, J1248	\$0.00	\$0.00
Wagner to J1401 138 kV	MTEP 17 Project 11884 in service 5/14/2021 raises limit to 288 MVA SN, 332 MVA SE	5/14/2021	42.83	0.4534			J1206, J1207, J1430	\$0.00	\$0.00
Monroe - Lalendorf 345kV	MTEP 20116 Raises limits to 1486 MVA SN, 1824 MVA SE	12/31/2024	44.54	0.1219			J1226, J1196, J1194, J1224, J1350, J1297, J1394, J1322, J1319, J1379, J1203, J1331, J1449, J1403, J1389, J1255, J1307, J1292, J1472, J1310, J1320, J1399, J1401, J1248, J1430, J1375, J1210, J1211, J1192, J1206, J1207, J1329, J1327	\$0.00	\$0.00
Cold Water to Ash Rd Jn 138 kV	MTEP 20 project 18084 Raises sag to 166 MVA	12/31/2023	47.21	0.1912			J1320, J1401, J1206, J1207	\$0.00	\$0.00
Ash Rd Jn to Moore Rd 138 kV	MTEP 20 project 18084 Raises sag to 166 MVA	12/31/2023	49.76	0.1912			J1320, J1401, J1206, J1207	\$0.00	\$0.00
Coldwater - J1320 138kV	Reuild 2 miles of 336 and 477 ACSR to 954 ACSR. (288 MVA SN, 332 MVA SE)		50	0.1155	yes	J1320	J1320, J1401, J1206, J1207	\$280,257.14	\$3,400,000.00

9.10. J1210

This generator is determined to be fully deliverable for 50 MW, contingent upon the system upgrades and contingent facilities identified in the NRIS analysis. Table 10 shows the NRIS results and cost estimates determined in the NRIS analysis.

Table 12: NRIS Results for J1210

J1210 Deliverable (NRIS) Amount in 2024 Case: (Conditional on ERIS and case assumptions)						42.83 MW (85.66%)			

Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Mitigation /Upgrade	MTEP project ISD	Level of Service Attainable (MW)	Distribution Factor	Constraint in ERI Analysis?	Projects Associated with ERIS Constraint	Projects Associated with NRIS Constraint	Upgrade Cost Allocated to Project	Total Cost of Upgrade (\$)
Monroe - Lalendorf 345kV	MTEP 20116 Raises limits to 1486 MVA SN, 1824 MVA SE	12/31/2024	50	0.1983			J1226, J1196, J1194, J1224, J1350, J1297, J1394, J1322, J1319, J1379, J1203, J1331, J1449, J1403, J1389, J1255, J1307, J1292, J1472, J1310, J1320, J1399, J1401, J1248, J1430, J1375, J1210, J1211, J1192, J1206, J1207, J1329, J1327	\$0.00	\$0.00

9.11. J1211

This generator is determined to be fully deliverable for 50 MW, contingent upon the system upgrades and contingent facilities identified in the NRIS analysis. Table 13 shows the NRIS results and cost estimates determined in the NRIS analysis.

Table 13: NRIS Results for J1211

J1211 Deliverable (NRIS) Amount in 2024 Case: (Conditional on ERIS and case assumptions)						0 MW (0%)			
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Mitigation /Upgrade	MTEP project ISD	Level of Service Attainable (MW)	Distribution Factor	Constraint in ERI Analysis?	Projects Associated with ERIS Constraint	Projects Associated with NRIS Constraint	Upgrade Cost Allocated to Project	Total Cost of Upgrade (\$)
Luce - Regal 138kV	Reconductor 0.75 miles of 336 ACSR with 795 ACSS. (288 MVA SN, 332 MVA SE)		42.83	0.2239			J1211	\$1,000,000.00	\$1,000,000.00
Monroe - Lalendorf 345kV	MTEP 20116 Raises limits to 1486 MVA SN, 1824 MVA SE	12/31/2024	45.84	0.1972			J1226, J1196, J1194, J1224, J1350, J1297, J1394, J1322, J1319, J1379, J1203, J1331, J1449, J1403, J1389, J1255, J1307, J1292, J1472, J1310, J1320, J1399, J1401, J1248, J1430, J1375, J1210, J1211, J1192, J1206, J1207, J1329, J1327	\$0.00	\$0.00
J1379 - Derby Junction 138kV	MTEP 21 Target A Project 20016 Rebuilds the 336 to 954 ACSR. (288 MVA SN, 332 MVA SE)	6/1/2025	48.31	0.0925	yes	J1379	J1379, J1211	\$0.00	\$0.00
Eureka - Derby Junction 138kV	MTEP 21 Target A Project 20016 Rebuilds the 336 to 954 ACSR. (288 MVA SN, 332 MVA SE)	6/1/2025	50	0.0925	yes	J1379	J1379, J1211	\$0.00	\$0.00

9.12. J1224

This generator is determined to be fully deliverable for 170 MW, contingent upon the system upgrades and contingent facilities identified in the NRIS analysis. Table 14 shows the NRIS results and cost estimates determined in the NRIS analysis.

Table 14: NRIS Results for J1224

J1224 Deliverable (NRIS) Amount in 2024 Case: (Conditional on ERIIS and case assumptions)						145.63 MW (85.66%)			
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Mitigation /Upgrade	MTEP project ISD	Level of Service Attainable (MW)	Distribution Factor	Constraint in ERIIS Analysis?	Projects Associated with ERIS Constraint	Projects Associated with NRIS Constraint	Upgrade Cost Allocated to Project	Total Cost of Upgrade (\$)
Monroe - Lalendorf 345kV	MTEP 20116 Raises limits to 1486 MVA SN, 1824 MVA SE	12/31/2024	170	0.2568			J1226, J1196, J1194, J1224, J1350, J1297, J1394, J1322, J1319, J1379, J1203, J1331, J1449, J1403, J1389, J1255, J1307, J1292, J1472, J1310, J1320, J1399, J1401, J1248, J1430, J1375, J1210, J1211, J1192, J1206, J1207, J1329, J1327	\$0.00	\$0.00

9.13. J1226

This generator is determined to be fully deliverable for 200 MW, contingent upon the system upgrades and contingent facilities identified in the NRIS analysis. Table 15 shows the NRIS results and cost estimates determined in the NRIS analysis.

Table 15: NRIS Results for J1226

J1226 Deliverable (NRIS) Amount in 2024 Case: (Conditional on ERIIS and case assumptions)						171.33 MW (85.66%)			
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Mitigation /Upgrade	MTEP project ISD	Level of Service Attainable (MW)	Distribution Factor	Constraint in ERIIS Analysis?	Projects Associated with ERIS Constraint	Projects Associated with NRIS Constraint	Upgrade Cost Allocated to Project	Total Cost of Upgrade (\$)
Monroe - Lalendorf 345kV	MTEP 20116 Raises limits to 1486 MVA SN, 1824 MVA SE	12/31/2024	200	0.2708			J1226, J1196, J1194, J1224, J1350, J1297, J1394, J1322, J1319, J1379, J1203, J1331, J1449, J1403, J1389, J1255, J1307, J1292, J1472, J1310, J1320, J1399, J1401, J1248, J1430, J1375, J1210, J1211, J1192, J1206, J1207, J1329, J1327	\$0.00	\$0.00

9.14. J1248

This generator is determined to be fully deliverable for 100 MW, contingent upon the system upgrades and contingent facilities identified in the NRIS analysis. Table 16 shows the NRIS results and cost estimates determined in the NRIS analysis.

Table 16: NRIS Results for J1248

J1248 Deliverable (NRIS) Amount in 2024 Case: (Conditional on ERIIS and case assumptions)					29.97 MW (29.97%)				
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Mitigation /Upgrade	MTEP project ISD	Level of Service Attainable (MW)	Distribution Factor	Constraint in ERIIS Analysis?	Projects Associated with ERIS Constraint	Projects Associated with NRIS Constraint	Upgrade Cost Allocated to Project	Total Cost of Upgrade (\$)
Batavia - J1401 138kV	MTEP 17 Project 11884 in service 5/14/2021 raises limit to 288 MVA SN, 332 MVA SE	5/14/2021	33.7	0.0505	yes	J1401, J1206, J1207	J1401, J1206, J1207, J1430, J1248	\$0.00	\$0.00
Cement City Junction - Seamless 138kV	MTEP 20 15939 increases the limit to 227 MVA SN, 286 MVA SE	12/31/2022	57.89	0.0609			J1472, J1310, J1399, J1248	\$0.00	\$0.00
J758 (Mud Lake) - Verona #2 138kV	Rebuild about 4.4 miles of 336 ACSR to 954 ACSR. (288 MVA SN, 332 MVA SE)		65.87	0.5112	yes	J1310, J1430, J1248	J1310, J1430, J1248, J1472	as per ERIIS	\$7,100,000.00
Wagner to J1401 138 kV	MTEP 17 Project 11884 in service 5/14/2021 raises limit to 288 MVA SN, 332 MVA SE	5/14/2021	85.67	0.0505			J1206, J1207, J1430	\$0.00	\$0.00
Monroe - Lalendorf 345kV	MTEP 20116 Raises limits to 1486 MVA SN, 1824 MVA SE	12/31/2024	100	0.1458			J1226, J1196, J1194, J1224, J1350, J1297, J1394, J1322, J1319, J1379, J1203, J1331, J1449, J1403, J1389, J1255, J1307, J1292, J1472, J1310, J1320, J1399, J1401, J1248, J1430, J1375, J1210, J1211, J1192, J1206, J1207, J1329, J1327	\$0.00	\$0.00

9.15. J1255

This generator is determined to be fully deliverable for 200 MW, contingent upon the system upgrades and contingent facilities identified in the NRIS analysis. Table 17 shows the NRIS results and cost estimates determined in the NRIS analysis.

Table 17: NRIS Results for J1255

J1255 Deliverable (NRIS) Amount in 2024 Case: (Conditional on ERIIS and case assumptions)					171.33 MW (85.66%)				
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Mitigation /Upgrade	MTEP project ISD	Level of Service Attainable (MW)	Distribution Factor	Constraint in ERIIS Analysis?	Projects Associated with ERIS Constraint	Projects Associated with NRIS Constraint	Upgrade Cost Allocated to Project	Total Cost of Upgrade (\$)

Monroe - Lalendorf 345kV	MTEP 20116 Raises limits to 1486 MVA SN, 1824 MVA SE	12/31/2024	173.08	0.1183			J1226, J1196, J1194, J1224, J1350, J1297, J1394, J1322, J1319, J1379, J1203, J1331, J1449, J1403, J1389, J1255, J1307, J1292, J1472, J1310, J1320, J1399, J1401, J1248, J1430, J1375, J1210, J1211, J1192, J1206, J1207, J1329, J1327	\$0.00	\$0.00
J1379 - Deja Junction 138kV	MTEP 21 Target A Project 20016 Rebuilds the 336 to 954 ACSR. (288 MVA SN, 332 MVA SE)	6/1/2025	200	0.0529	yes	J1379, J1403, J1403	J1379, J1403, J1255, J1327	\$0.00	\$0.00

9.16. J1292

This generator is determined to be fully deliverable for 198 MW, contingent upon the system upgrades and contingent facilities identified in the NRIS analysis. Table 18 shows the NRIS results and cost estimates determined in the NRIS analysis.

Table 18: NRIS Results for J1292

J1292 Deliverable (NRIS) Amount in 2024 Case: (Conditional on ERIS and case assumptions)						169.62 MW (85.67%)			
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Mitigation /Upgrade	MTEP project ISD	Level of Service Attainable (MW)	Distribution Factor	Constraint in ERIS Analysis?	Projects Associated with ERIS Constraint	Projects Associated with NRIS Constraint	Upgrade Cost Allocated to Project	Total Cost of Upgrade (\$)
Monroe - Lalendorf 345kV	MTEP 20116 Raises limits to 1486 MVA SN, 1824 MVA SE	12/31/2024	198	0.1145			J1226, J1196, J1194, J1224, J1350, J1297, J1394, J1322, J1319, J1379, J1203, J1331, J1449, J1403, J1389, J1255, J1307, J1292, J1472, J1310, J1320, J1399, J1401, J1248, J1430, J1375, J1210, J1211, J1192, J1206, J1207, J1329, J1327	\$0.00	\$0.00

9.17. J1297

This generator is determined to be fully deliverable for 170 MW, contingent upon the system upgrades and contingent facilities identified in the NRIS analysis. Table 19 shows the NRIS results and cost estimates determined in the NRIS analysis.

Table 19: NRIS Results for J1297

J1297 Deliverable (NRIS) Amount in 2024 Case: (Conditional on ERIS and case assumptions)						74.01 MW (43.54%)			

Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Mitigation /Upgrade	MTEP project ISD	Level of Service Attainable (MW)	Distribution Factor	Constraint in ERI Analysis?	Projects Associated with ERIS Constraint	Projects Associated with NRIS Constraint	Upgrade Cost Allocated to Project	Total Cost of Upgrade (\$)
Goss - Pasadena Junciton 138kV	Raise the sag limit of 2.2 milies of 795 ACSR to 280 MVA		105.75	0.1182			J1297, J1394	\$0.00	\$0.00
Cornell - North Corunna Junction 138kV	Above Conductor Max. Rebuild 3 miles of 477 to 954 ACSR. (288 MVA SN, 332 MVA SE)		145.63	0.3195			J1297	\$4,900,000.00	\$4,900,000.00
Monroe - Lalendorf 345kV	MTEP 20116 Raises limits to 1486 MVA SN, 1824 MVA SE	12/31/2024	170	0.2049			J1226, J1196, J1194, J1224, J1350, J1297, J1394, J1322, J1319, J1379, J1203, J1331, J1449, J1403, J1389, J1255, J1307, J1292, J1472, J1310, J1320, J1399, J1401, J1248, J1430, J1375, J1210, J1211, J1192, J1206, J1207, J1329, J1327	\$0.00	\$0.00

9.18. J1307

This generator is determined to be fully deliverable for 150 MW, contingent upon the system upgrades and contingent facilities identified in the NRIS analysis. Table 20 shows the NRIS results and cost estimates determined in the NRIS analysis.

Table 20: NRIS Results for J1307

J1307 Deliverable (NRIS) Amount in 2024 Case: (Conditional on ERI and case assumptions)						128.5 MW (85.67%)			
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Mitigation /Upgrade	MTEP project ISD	Level of Service Attainable (MW)	Distribution Factor	Constraint in ERI Analysis?	Projects Associated with ERIS Constraint	Projects Associated with NRIS Constraint	Upgrade Cost Allocated to Project	Total Cost of Upgrade (\$)
Monroe - Lalendorf 345kV	MTEP 20116 Raises limits to 1486 MVA SN, 1824 MVA SE	12/31/2024	150	0.1527			J1226, J1196, J1194, J1224, J1350, J1297, J1394, J1322, J1319, J1379, J1203, J1331, J1449, J1403, J1389, J1255, J1307, J1292, J1472, J1310, J1320, J1399, J1401, J1248, J1430, J1375, J1210, J1211, J1192, J1206, J1207, J1329, J1327	\$0.00	\$0.00

9.19. J1310



This generator is determined to be fully deliverable for 125 MW, contingent upon the system upgrades and contingent facilities identified in the NRIS analysis. Table 21 shows the NRIS results and cost estimates determined in the NRIS analysis.

Table 21: NRIS Results for J1310

J1310 Deliverable (NRIS) Amount in 2024 Case: (Conditional on ERIS and case assumptions)					0 MW (0%)				
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Mitigation /Upgrade	MTEP project ISD	Level of Service Attainable (MW)	Distribution Factor	Constraint in ERIS Analysis?	Projects Associated with ERIS Constraint	Projects Associated with NRIS Constraint	Upgrade Cost Allocated to Project	Total Cost of Upgrade (\$)
Leoni - Washington Junction 138kV	Rebuild about 13 miles of 477 ACSR to 954 ACSR. (288 MVA SN, 331 MVA SE)		42.13	0.065	yes	J1472, J1310	J1472, J1310	as per ERIS	\$21,000,000.00
Cement City Junction - Seamless 138kV	MTEP 20 15939 increases the limit to 227 MVA SN, 286 MVA SE	12/31/2022	72.36	0.1088			J1472, J1310, J1399, J1248	\$0.00	\$0.00
J758 (Mud Lake) - Verona #2 138kV	Rebuild about 4.4 miles of 336 ACSR to 954 ACSR. (288 MVA SN, 332 MVA SE)		93.66	0.5063	yes	J1310, J1430, J1248	J1310, J1430, J1248, J1472	as per ERIS	\$7,100,000.00
Lark - Washington Junction 138kV	Raise sag limit to 195 MVA		107.08	0.0688	yes	J1472, J1310	J1472, J1310, J1399	\$0.00	\$0.00
Monroe - Lalendorf 345kV	MTEP 20116 Raises limits to 1486 MVA SN, 1824 MVA SE	12/31/2024	125	0.1711			J1226, J1196, J1194, J1224, J1350, J1297, J1394, J1322, J1319, J1379, J1203, J1331, J1449, J1403, J1389, J1255, J1307, J1292, J1472, J1310, J1320, J1399, J1401, J1248, J1430, J1375, J1210, J1211, J1192, J1206, J1207, J1329, J1327	\$0.00	\$0.00

9.20. J1319

This generator is determined to be fully deliverable for 117.5 MW, contingent upon the system upgrades and contingent facilities identified in the NRIS analysis. Table 22 shows the NRIS results and cost estimates determined in the NRIS analysis.

Table 22: NRIS Results for J1319

J1319 Deliverable (NRIS) Amount in 2024 Case: (Conditional on ERIS and case assumptions)					49.05 MW (41.74%)				
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Mitigation /Upgrade	MTEP project ISD	Level of Service Attainable (MW)	Distribution Factor	Constraint in ERIS Analysis?	Projects Associated with ERIS Constraint	Projects Associated with NRIS Constraint	Upgrade Cost Allocated to Project	Total Cost of Upgrade (\$)
Caniff 120kV phase shifter	Can be fixed by operating phase shifter		64.89	0.06			J1196, J1319, J1331, J1203	\$0.00	\$0.00
Caniff 345 / 120kV transformer	Can be fixed by operating phase shifter		100.66	0.06			J1196, J1319, J1331, J1203	\$0.00	\$0.00

Monroe - Lalendorf 345kV	MTEP 20116 Raises limits to 1486 MVA SN, 1824 MVA SE	12/31/2024	117.5	0.2534		J1226, J1196, J1194, J1224, J1350, J1297, J1394, J1322, J1319, J1379, J1203, J1331, J1449, J1403, J1389, J1255, J1307, J1292, J1472, J1310, J1320, J1399, J1401, J1248, J1430, J1375, J1210, J1211, J1192, J1206, J1207, J1329, J1327	\$0.00	\$0.00
--------------------------	--	------------	-------	--------	--	---	--------	--------

9.21. J1320

This generator is determined to be fully deliverable for 150 MW, contingent upon the system upgrades and contingent facilities identified in the NRIS analysis. Table 23 shows the NRIS results and cost estimates determined in the NRIS analysis.

Table 23: NRIS Results for J1320

J1320 Deliverable (NRIS) Amount in 2024 Case: (Conditional on ERIS and case assumptions)						102.31 MW (68.21%)			
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Mitigation /Upgrade	MTEP project ISD	Level of Service Attainable (MW)	Distribution Factor	Constraint in ERIS Analysis?	Projects Associated with ERIS Constraint	Projects Associated with NRIS Constraint	Upgrade Cost Allocated to Project	Total Cost of Upgrade (\$)
J1401 - Wagner 138kV	MTEP 17 Project 11884 in service 5/14/2021 raises limit to 288 MVA SN, 332 MVA SE	5/14/2021	128.5	0.4811	yes	J1401, J1320	J1401, J1320	\$0.00	\$0.00
Monroe - Lalendorf 345kV	MTEP 20116 Raises limits to 1486 MVA SN, 1824 MVA SE	12/31/2024	133.61	0.1213			J1226, J1196, J1194, J1224, J1350, J1297, J1394, J1322, J1319, J1379, J1203, J1331, J1449, J1403, J1389, J1255, J1307, J1292, J1472, J1310, J1320, J1399, J1401, J1248, J1430, J1375, J1210, J1211, J1192, J1206, J1207, J1329, J1327	\$0.00	\$0.00
Cold Water to Ash Rd Jn 138 kV	MTEP 20 project 18084 Raises sag to 166 MVA	12/31/2023	137.7	0.2059			J1320, J1401, J1206, J1207	\$0.00	\$0.00
Coldwater - J1320 138kV	Reuild 2 miles of 336 and 477 ACSR to 954 ACSR. (288 MVA SN, 332 MVA SE)		141.64	0.239	yes	J1320	J1320, J1401, J1206, J1207	as per ERIS	\$3,400,000.00
Ash RdJn to Moore Rd 138 kV	MTEP 20 project 18084 Raises sag to 166 MVA	12/31/2023	150	0.2059			J1320, J1401, J1206, J1207	\$0.00	\$0.00

9.22. J1322

This generator is determined to be fully deliverable for 200 MW, contingent upon the system upgrades and contingent facilities identified in the NRIS analysis. Table 24 shows the NRIS results and cost estimates determined in the NRIS analysis.

Table 24: NRIS Results for J1322

J1322 Deliverable (NRIS) Amount in 2024 Case: (Conditional on ERIIS and case assumptions)						171.33 MW (85.66%)			
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Mitigation /Upgrade	MTEP project ISD	Level of Service Attainable (MW)	Distribution Factor	Constraint in ERIIS Analysis?	Projects Associated with ERIS Constraint	Projects Associated with NRIS Constraint	Upgrade Cost Allocated to Project	Total Cost of Upgrade (\$)
Monroe - Lalendorf 345kV	MTEP 20116 Raises limits to 1486 MVA SN, 1824 MVA SE	12/31/2024	200	0.1525			J1226, J1196, J1194, J1224, J1350, J1297, J1394, J1322, J1319, J1379, J1203, J1331, J1449, J1403, J1389, J1255, J1307, J1292, J1472, J1310, J1320, J1399, J1401, J1248, J1430, J1375, J1210, J1211, J1192, J1206, J1207, J1329, J1327	\$0.00	\$0.00

9.23. J1327

This generator is determined to be fully deliverable for 20 MW, contingent upon the system upgrades and contingent facilities identified in the NRIS analysis. Table 25 shows the NRIS results and cost estimates determined in the NRIS analysis.

Table 25: NRIS Results for J1327

J1327 Deliverable (NRIS) Amount in 2024 Case: (Conditional on ERIIS and case assumptions)						17.13 MW (85.65%)			
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Mitigation /Upgrade	MTEP project ISD	Level of Service Attainable (MW)	Distribution Factor	Constraint in ERIIS Analysis?	Projects Associated with ERIS Constraint	Projects Associated with NRIS Constraint	Upgrade Cost Allocated to Project	Total Cost of Upgrade (\$)
Monroe - Lalendorf 345kV	MTEP 20116 Raises limits to 1486 MVA SN, 1824 MVA SE	12/31/2024	17.31	0.1183			J1226, J1196, J1194, J1224, J1350, J1297, J1394, J1322, J1319, J1379, J1203, J1331, J1449, J1403, J1389, J1255, J1307, J1292, J1472, J1310, J1320, J1399, J1401, J1248, J1430, J1375, J1210, J1211, J1192, J1206, J1207, J1329, J1327	\$0.00	\$0.00

J1379 - Deja Junction 138kV	MTEP 21 Target A Project 20016 Rebuilds the 336 to 954 ACSR. (288 MVA SN, 332 MVA SE)	6/1/2025	20	0.0529	yes	J1379, J1403, J1403	J1379, J1403, J1255, J1327	\$0.00	\$0.00
-----------------------------	---	----------	----	--------	-----	---------------------	----------------------------	--------	--------

9.24. J1329

This generator is determined to be fully deliverable for 20 MW, contingent upon the system upgrades and contingent facilities identified in the NRIS analysis. Table 26 shows the NRIS results and cost estimates determined in the NRIS analysis.

Table 26: NRIS Results for J1329

J1329 Deliverable (NRIS) Amount in 2024 Case: (Conditional on ERS and case assumptions)						17.13 MW (85.65%)			
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Mitigation / Upgrade	MTEP project ISD	Level of Service Attainable (MW)	Distribution Factor	Constraint in ERS Analysis?	Projects Associated with ERS Constraint	Projects Associated with NRIS Constraint	Upgrade Cost Allocated to Project	Total Cost of Upgrade (\$)
Monroe - Lalendorf 345kV	MTEP 20116 Raises limits to 1486 MVA SN, 1824 MVA SE	12/31/2024	20	0.2568			J1226, J1196, J1194, J1224, J1350, J1297, J1394, J1322, J1319, J1379, J1203, J1331, J1449, J1403, J1389, J1255, J1307, J1292, J1472, J1310, J1320, J1399, J1401, J1248, J1430, J1375, J1210, J1211, J1192, J1206, J1207, J1329, J1327	\$0.00	\$0.00

9.25. J1331

This generator is determined to be fully deliverable for 100 MW, contingent upon the system upgrades and contingent facilities identified in the NRIS analysis. Table 27 shows the NRIS results and cost estimates determined in the NRIS analysis.

Table 27: NRIS Results for J1331

J1331 Deliverable (NRIS) Amount in 2024 Case: (Conditional on ERS and case assumptions)						41.75 MW (41.75%)			
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Mitigation / Upgrade	MTEP project ISD	Level of Service Attainable (MW)	Distribution Factor	Constraint in ERS Analysis?	Projects Associated with ERS Constraint	Projects Associated with NRIS Constraint	Upgrade Cost Allocated to Project	Total Cost of Upgrade (\$)
Caniff 120kV phase shifter	Can be fixed by operating phase shifter		55.23	0.0684			J1196, J1319, J1331, J1203	\$0.00	\$0.00
Caniff 345 / 120kV transformer	Can be fixed by operating phase shifter		85.67	0.0684			J1196, J1319, J1331, J1203	\$0.00	\$0.00

Monroe - Lalendorf 345kV	MTEP 20116 Raises limits to 1486 MVA SN, 1824 MVA SE	12/31/2024	100	0.2701		J1226, J1196, J1194, J1224, J1350, J1297, J1394, J1322, J1319, J1379, J1203, J1331, J1449, J1403, J1389, J1255, J1307, J1292, J1472, J1310, J1320, J1399, J1401, J1248, J1430, J1375, J1210, J1211, J1192, J1206, J1207, J1329, J1327	\$0.00	\$0.00
--------------------------	---	------------	-----	--------	--	---	--------	--------

9.26. J1350

This generator is determined to be fully deliverable for 150 MW, contingent upon the system upgrades and contingent facilities identified in the NRIS analysis. Table 28 shows the NRIS results and cost estimates determined in the NRIS analysis.

Table 28: NRIS Results for J1350

J1350 Deliverable (NRIS) Amount in 2024 Case: (Conditional on ERI and case assumptions)						128.5 MW (85.67%)			
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Mitigation / Upgrade	MTEP project ISD	Level of Service Attainable (MW)	Distribution Factor	Constraint in ERI Analysis?	Projects Associated with ERI Constraint	Projects Associated with NRIS Constraint	Upgrade Cost Allocated to Project	Total Cost of Upgrade (\$)
Monroe - Lalendorf 345kV	MTEP 20116 Raises limits to 1486 MVA SN, 1824 MVA SE	12/31/2024	150	0.2723			J1226, J1196, J1194, J1224, J1350, J1297, J1394, J1322, J1319, J1379, J1203, J1331, J1449, J1403, J1389, J1255, J1307, J1292, J1472, J1310, J1320, J1399, J1401, J1248, J1430, J1375, J1210, J1211, J1192, J1206, J1207, J1329, J1327	\$0.00	\$0.00

9.27. J1375

This generator is determined to be fully deliverable for 100 MW, contingent upon the system upgrades and contingent facilities identified in the NRIS analysis. Table 29 shows the NRIS results and cost estimates determined in the NRIS analysis.

Table 29: NRIS Results for J1375

J1375 Deliverable (NRIS) Amount in 2024 Case:	85.67 MW (85.67%)
---	-------------------

(Conditional on ERS and case assumptions)									
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Mitigation /Upgrade	MTEP project ISD	Level of Service Attainable (MW)	Distribution Factor	Constraint in ERS Analysis?	Projects Associated with ERS Constraint	Projects Associated with NRIS Constraint	Upgrade Cost Allocated to Project	Total Cost of Upgrade (\$)
Monroe - Lalendorf 345kV	MTEP 20116 Raises limits to 1486 MVA SN, 1824 MVA SE	12/31/2024	95.25	0.1105			J1226, J1196, J1194, J1224, J1350, J1297, J1394, J1322, J1319, J1379, J1203, J1331, J1449, J1403, J1389, J1255, J1307, J1292, J1472, J1310, J1320, J1399, J1401, J1248, J1430, J1375, J1210, J1211, J1192, J1206, J1207, J1329, J1327	\$0.00	\$0.00
Benston Junction - Peterson Junction 138kV	Raise the sag limit of the 336 ACSR to 111 MVA		95.83	0.5636			J1375	\$0.00	\$0.00
Cobb - Peterson Junction 138kV	Raise the sag limit of the 336 ACSR to 111 MVA		100	0.5636			J1375	\$0.00	\$0.00

9.28. J1379

This generator is determined to be fully deliverable for 200 MW, contingent upon the system upgrades and contingent facilities identified in the NRIS analysis. Table 30 shows the NRIS results and cost estimates determined in the NRIS analysis.

Table 30: NRIS Results for J1379

J1379 Deliverable (NRIS) Amount in 2024 Case: (Conditional on ERS and case assumptions)					149.31 MW (74.66%)				
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Mitigation /Upgrade	MTEP project ISD	Level of Service Attainable (MW)	Distribution Factor	Constraint in ERS Analysis?	Projects Associated with ERS Constraint	Projects Associated with NRIS Constraint	Upgrade Cost Allocated to Project	Total Cost of Upgrade (\$)
Luce - Bluegrass Junction 138kV	Reconductor 11.1 miles of 336 ACSR with 795ACSS. (416MVA SN, 416 MVA SE)		168.39	0.1799			J1379	\$5,000,000.00	\$5,000,000.00
Summerton - Bluegrass Junction 138kV	Reconductor 11.1 miles of 336 ACSR with 795ACSS. (416MVA SN, 416 MVA SE)		170.23	0.1799			J1379	\$5,000,000.00	\$5,000,000.00
J1379 - Derby Junction 138kV	MTEP 21 Target A Project 20016 Rebuilds the 336 to 954 ACSR. (288 MVA SN, 332 MVA SE)		170.7	1	yes	J1379	J1379, J1211	\$0.00	\$0.00
J1379 - Deja Junction 138kV	MTEP 21 Target A Project 20016 Rebuilds the 336 to 954 ACSR. (288 MVA SN, 332 MVA SE)	6/1/2025	171.33	1	yes	J1379, J1403, J1403	J1379, J1403, J1255, J1327	\$0.00	\$0.00

Monroe - Lalendorf 345kV	MTEP 20116 Raises limits to 1486 MVA SN, 1824 MVA SE	12/31/2024	176.39	0.1445			J1226, J1196, J1194, J1224, J1350, J1297, J1394, J1322, J1319, J1379, J1203, J1331, J1449, J1403, J1389, J1255, J1307, J1292, J1472, J1310, J1320, J1399, J1401, J1248, J1430, J1375, J1210, J1211, J1192, J1206, J1207, J1329, J1327	\$0.00	\$0.00
Eureka - Derby Junction 138kV	MTEP 21 Target A Project 20016 Rebuilds the 336 to 954 ACSR. (288 MVA SN, 332 MVA SE)		200	1	yes	J1379	J1379, J1211	\$0.00	\$0.00

9.29. J1389

This generator is determined to be fully deliverable for 120 MW, contingent upon the system upgrades and contingent facilities identified in the NRIS analysis. Table 31 shows the NRIS results and cost estimates determined in the NRIS analysis.

Table 31: NRIS Results for J1389

J1389 Deliverable (NRIS) Amount in 2024 Case: (Conditional on ERS and case assumptions)						102.8 MW (85.67%)			
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Mitigation / Upgrade	MTEP project ISD	Level of Service Attainable (MW)	Distribution Factor	Constraint in ERS Analysis?	Projects Associated with ERS Constraint	Projects Associated with NRIS Constraint	Upgrade Cost Allocated to Project	Total Cost of Upgrade (\$)
Monroe - Lalendorf 345kV	MTEP 20116 Raises limits to 1486 MVA SN, 1824 MVA SE	12/31/2024	120	0.1989			J1226, J1196, J1194, J1224, J1350, J1297, J1394, J1322, J1319, J1379, J1203, J1331, J1449, J1403, J1389, J1255, J1307, J1292, J1472, J1310, J1320, J1399, J1401, J1248, J1430, J1375, J1210, J1211, J1192, J1206, J1207, J1329, J1327	\$0.00	\$0.00

9.30. J1394

This generator is determined to be fully deliverable for 150 MW, contingent upon the system upgrades and contingent facilities identified in the NRIS analysis. Table 32 shows the NRIS results and cost estimates determined in the NRIS analysis.

Table 32: NRIS Results for J1394

J1394 Deliverable (NRIS) Amount in 2024 Case:	58.64 MW (39.09%)
---	-------------------

(Conditional on ERIIS and case assumptions)									
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Mitigation /Upgrade	MTEP project ISD	Level of Service Attainable (MW)	Distribution Factor	Constraint in ERIIS Analysis?	Projects Associated with ERIS Constraint	Projects Associated with NRIS Constraint	Upgrade Cost Allocated to Project	Total Cost of Upgrade (\$)
J1088 - Layton 138kV	Rebuild about 6.9 miles of 477 ACSR to 954ACSR. (288 MVA SN, 332 MVA SE)		58.9	0.9989	yes	J1394	J1394	as per ERIIS	\$11,100,000.00
Cornell - J1088 138kV	Rebuild about 8.5miles of 477 ACSR to 954ACSR. (288 MVA SN, 332 MVA SE)		65.31	0.9989	yes	J1394	J1394	as per ERIIS	\$13,700,000.00
Goss - Pasadena Junciton 138kV	Raise the sag limit of 2.2 milies of 795 ACSR to 280 MVA		128.5	0.062			J1297, J1394	\$0.00	\$0.00
Monroe - Lalendorf 345kV	MTEP 20116 Raises limits to 1486 MVA SN, 1824 MVA SE	12/31/2024	150	0.2042			J1226, J1196, J1194, J1224, J1350, J1297, J1394, J1322, J1319, J1379, J1203, J1331, J1449, J1403, J1389, J1255, J1307, J1292, J1472, J1310, J1320, J1399, J1401, J1248, J1430, J1375, J1210, J1211, J1192, J1206, J1207, J1329, J1327	\$0.00	\$0.00

9.31. J1399

This generator is determined to be fully deliverable for 90 MW, contingent upon the system upgrades and contingent facilities identified in the NRIS analysis. Table 33 shows the NRIS results and cost estimates determined in the NRIS analysis.

Table 33: NRIS Results for J1399

J1399 Deliverable (NRIS) Amount in 2024 Case:					30.33 MW (33.7%)				
(Conditional on ERIIS and case assumptions)									
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Mitigation /Upgrade	MTEP project ISD	Level of Service Attainable (MW)	Distribution Factor	Constraint in ERIIS Analysis?	Projects Associated with ERIS Constraint	Projects Associated with NRIS Constraint	Upgrade Cost Allocated to Project	Total Cost of Upgrade (\$)
Cement City Junction - Seamless 138kV	MTEP 20 15939 increases the limit to 227 MVA SN, 286 MVA SE	12/31/2022	67.44	0.0865			J1472, J1310, J1399, J1248	\$0.00	\$0.00
Lark - Washington Junction 138kV	Raise sag limit to 195 MVA		77.1	0.0521	yes	J1472, J1310	J1472, J1310, J1399	\$0.00	\$0.00

Monroe - Lalendorf 345kV	MTEP 20116 Raises limits to 1486 MVA SN, 1824 MVA SE	12/31/2024	90	0.1941		J1226, J1196, J1194, J1224, J1350, J1297, J1394, J1322, J1319, J1379, J1203, J1331, J1449, J1403, J1389, J1255, J1307, J1292, J1472, J1310, J1320, J1399, J1401, J1248, J1430, J1375, J1210, J1211, J1192, J1206, J1207, J1329, J1327	\$0.00	\$0.00
--------------------------	--	------------	----	--------	--	---	--------	--------

9.32. J1401

This generator is determined to be fully deliverable for 150 MW, contingent upon the system upgrades and contingent facilities identified in the NRIS analysis. Table 34 shows the NRIS results and cost estimates determined in the NRIS analysis.

Table 34: NRIS Results for J1401

J1401 Deliverable (NRIS) Amount in 2024 Case: (Conditional on ERIS and case assumptions)						44.96 MW (29.97%)			
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Mitigation /Upgrade	MTEP project ISD	Level of Service Attainable (MW)	Distribution Factor	Constraint in ERIS Analysis?	Projects Associated with ERIS Constraint	Projects Associated with NRIS Constraint	Upgrade Cost Allocated to Project	Total Cost of Upgrade (\$)
Batavia - J1401 138kV	MTEP 17 Project 11884 in service 5/14/2021 raises limit to 288 MVA SN, 332 MVA SE	5/14/2021	102.31	0.582	yes	J1401, J1206, J1207	J1401, J1206, J1207, J1430, J1248	\$0.00	\$0.00
J1401 - Wagner 138kV	MTEP 17 Project 11884 in service 5/14/2021 raises limit to 288 MVA SN, 332 MVA SE	5/14/2021	128.5	0.5277	yes	J1401, J1320	J1401, J1320	\$0.00	\$0.00
Monroe - Lalendorf 345kV	MTEP 20116 Raises limits to 1486 MVA SN, 1824 MVA SE	12/31/2024	133.61	0.1141			J1226, J1196, J1194, J1224, J1350, J1297, J1394, J1322, J1319, J1379, J1203, J1331, J1449, J1403, J1389, J1255, J1307, J1292, J1472, J1310, J1320, J1399, J1401, J1248, J1430, J1375, J1210, J1211, J1192, J1206, J1207, J1329, J1327	\$0.00	\$0.00
Cold Water to Ash Rd Jn 138 kV	MTEP 20 project 18084 Raises sag to 166 MVA	12/31/2023	137.7	0.1478			J1320, J1401, J1206, J1207	\$0.00	\$0.00
Coldwater - J1320 138kV	Reuild 2 miles of 336 and 477 ACSR to 954 ACSR. (288 MVA SN, 332 MVA SE)		141.64	0.1479	yes	J1320	J1320, J1401, J1206, J1207	\$1,077,800.00	\$3,400,000.00

9.33. J1403

This generator is determined to be fully deliverable for 200 MW, contingent upon the system upgrades and contingent facilities identified in the NRIS analysis. Table 35 shows the NRIS results and cost estimates determined in the NRIS analysis.

Table 35: NRIS Results for J1403

J1403 Deliverable (NRIS) Amount in 2024 Case: (Conditional on ERIS and case assumptions)						171.33 MW (85.66%)			
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Mitigation /Upgrade	MTEP project ISD	Level of Service Attainable (MW)	Distribution Factor	Constraint in ERIS Analysis?	Projects Associated with ERIS Constraint	Projects Associated with NRIS Constraint	Upgrade Cost Allocated to Project	Total Cost of Upgrade (\$)
Monroe - Lalendorf 345kV	MTEP 20116 Raises limits to 1486 MVA SN, 1824 MVA SE	12/31/2024	173.08	0.1286			J1226, J1196, J1194, J1224, J1350, J1297, J1394, J1322, J1319, J1379, J1203, J1331, J1449, J1403, J1389, J1255, J1307, J1292, J1472, J1310, J1320, J1399, J1401, J1248, J1430, J1375, J1210, J1211, J1192, J1206, J1207, J1329, J1327	\$0.00	\$0.00
J1379 - Deja Junction 138kV	MTEP 21 Target A Project 20016 Rebuilds the 336 to 954 ACSR. (288 MVA SN, 332 MVA SE)	6/1/2025	200	0.0791	yes	J1379, J1403, J1403	J1379, J1403, J1255, J1327	\$0.00	\$0.00

9.34. J1430

This generator is determined to be fully deliverable for 100 MW, contingent upon the system upgrades and contingent facilities identified in the NRIS analysis. Table 36 shows the NRIS results and cost estimates determined in the NRIS analysis.

Table 36: NRIS Results for J1430

J1430 Deliverable (NRIS) Amount in 2024 Case: (Conditional on ERIS and case assumptions)						0 MW (0%)			
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Mitigation /Upgrade	MTEP project ISD	Level of Service Attainable (MW)	Distribution Factor	Constraint in ERIS Analysis?	Projects Associated with ERIS Constraint	Projects Associated with NRIS Constraint	Upgrade Cost Allocated to Project	Total Cost of Upgrade (\$)
Wagner - Alder Creek Junction 138kV	MTEP 17 Project 11884 in service 5/14/2021 upgrades conductor to 954 ACSR. METC will need to upgrade station equipment to raise limit to 288 MVA SN, 332 MVA SE	5/14/2021	0	0.0604			J1430	\$100,000.00	\$100,000.00

Alder Creek - Barnum Junction 138kV	MTEP 17 Project 11884 in service 5/14/2021 upgrades conductor to 954 ACSR. METC will need to upgrade station equipment to raise limit to 288 MVA SN, 332 MVA SE	5/14/2021	29.97	0.0604			J1430	\$100,000.00	\$100,000.00
Batavia - J1401 138kV	MTEP 17 Project 11884 in service 5/14/2021 raises limit to 288 MVA SN, 332 MVA SE	5/14/2021	57.89	0.0623	yes	J1401, J1206, J1207	J1401, J1206, J1207, J1430, J1248	\$0.00	\$0.00
J758 (Mud Lake) - Verona #2 138kV	Rebuild about 4.4 miles of 336 ACSR to 954 ACSR. (288 MVA SN, 332 MVA SE)		65.87	0.5689	yes	J1310, J1430, J1248	J1310, J1430, J1248, J1472	as per ERIS	\$7,100,000.00
Wagner to J1401 138 kV	MTEP 17 Project 11884 in service 5/14/2021 raises limit to 288 MVA SN, 332 MVA SE	12/31/2024	85.67	0.0623			J1206, J1207, J1430	\$0.00	\$0.00
Monroe - Lalendorf 345kV	MTEP 20116 Raises limits to 1486 MVA SN, 1824 MVA SE	12/31/2024	100	0.1327			J1226, J1196, J1194, J1224, J1350, J1297, J1394, J1322, J1319, J1379, J1203, J1331, J1449, J1403, J1389, J1255, J1307, J1292, J1472, J1310, J1320, J1399, J1401, J1248, J1430, J1375, J1210, J1211, J1192, J1206, J1207, J1329, J1327	\$0.00	\$0.00

9.35. J1449

This generator is determined to be fully deliverable for 100 MW, contingent upon the system upgrades and contingent facilities identified in the NRIS analysis. Table 37 Table 23 shows the NRIS results and cost estimates determined in the NRIS analysis.

Table 37: NRIS Results for J1449

J1449 Deliverable (NRIS) Amount in 2024 Case: (Conditional on ERIS and case assumptions)						85.67 MW (85.67%)			
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Mitigation / Upgrade	MTEP project ISD	Level of Service Attainable (MW)	Distribution Factor	Constraint in ERIS Analysis?	Projects Associated with ERIS Constraint	Projects Associated with NRIS Constraint	Upgrade Cost Allocated to Project	Total Cost of Upgrade (\$)
Monroe - Lalendorf 345kV	MTEP 20116 Raises limits to 1486 MVA SN, 1824 MVA SE	12/31/2024	100	0.2596			J1226, J1196, J1194, J1224, J1350, J1297, J1394, J1322, J1319, J1379, J1203, J1331, J1449, J1403, J1389, J1255, J1307, J1292, J1472, J1310, J1320, J1399, J1401, J1248, J1430, J1375, J1210, J1211, J1192, J1206, J1207, J1329, J1327	\$0.00	\$0.00

9.36. J1472

This generator is determined to be fully deliverable for 100 MW, contingent upon the system upgrades and contingent facilities identified in the NRIS analysis. Table 38 shows the NRIS results and cost estimates determined in the NRIS analysis.

Table 38: NRIS Results for J1472

J1472 Deliverable (NRIS) Amount in 2024 Case: (Conditional on ERIS and case assumptions)					0 MW (0%)				
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Mitigation /Upgrade	MTEP project ISD	Level of Service Attainable (MW)	Distribution Factor	Constraint in ERIS Analysis?	Projects Associated with ERIS Constraint	Projects Associated with NRIS Constraint	Upgrade Cost Allocated to Project	Total Cost of Upgrade (\$)
Leoni - Washington Junction 138kV	Rebuild about 13 miles of 477 ACSR to 954 ACSR. (288 MVA SN, 331 MVA SE)		33.7	0.0935	yes	J1472, J1310	J1472, J1310	as per ERIS	\$21,000,000.00
Cement City Junction - Seamless 138kV	MTEP 20 15939 increases the limit to 227 MVA SN, 286 MVA SE	12/31/2022	57.65	0.1539			J1472, J1310, J1399, J1248	\$0.00	\$0.00
J758 (Mud Lake) - Verona #2 138kV	Rebuild about 4.4 miles of 336 ACSR to 954 ACSR. (288 MVA SN, 332 MVA SE)		74.93	0.0905	yes	J1310, J1430, J1248	J1310, J1430, J1248, J1472	\$354,686.47	\$7,100,000.00
Lark - Washington Junction 138kV	Raise sag limit to 195 MVA		85.67	0.1049	yes	J1472, J1310	J1472, J1310, J1399	\$0.00	\$0.00
Monroe - Lalendorf 345kV	MTEP 20116 Raises limits to 1486 MVA SN, 1824 MVA SE	12/31/2024	96.24	0.215			J1226, J1196, J1194, J1224, J1350, J1297, J1394, J1322, J1319, J1379, J1203, J1331, J1449, J1403, J1389, J1255, J1307, J1292, J1472, J1310, J1320, J1399, J1401, J1248, J1430, J1375, J1210, J1211, J1192, J1206, J1207, J1329, J1327	\$0.00	\$0.00

10. Shared Network Upgrades Analysis

Shared Network Upgrade (SNU) Analysis tests for Network Upgrades driven by higher queued interconnection projects was performed for this System Impact Study. No SNU was identified for this cycle.

11. Cost Allocation

The cost allocation of Network Upgrades for the study group reflects responsibilities for mitigating system impacts based on Interconnection Customer-elected level of Network Resource Interconnection service as of the draft System Impact Study report date.

11.1. Cost Assumptions for Network Upgrades

The cost estimate for each network upgrade identified in System Impact Study was provided by the corresponding transmission owning company.

11.2. Cost Allocation Methodology

The costs of Network Upgrades (NU) for a set of generation projects (one or more sub-groups or entire group with identified NU) are allocated based on the MW impact from each project on the constrained facilities in the Study Case.

Cost Allocation Methodology for Thermal Constraints

1. With all Study Group generation projects dispatched in the Post Case, all thermal constraints are identified.
2. Distribution factor from each project on each constraint is obtained.
3. For each thermal constraint, the maximum MW contribution (increasing flow) from each project is then calculated in the Post Case without any network upgrades.
4. For each thermal constraint, the cost estimates for one or a subset of NU are provided by the corresponding Transmission Owner.
5. Then the cost of each NU is allocated based on the pro rata share of the MW contribution from each project on the constraints mitigated or partly mitigated by this NU. The methodology to determine the cost allocation of one NU is:

$$\text{Cost of NU} = \frac{\text{Project A cost portion of NU} \times \text{Max(Proj.A MW contribution on constraint)}}{\sum_i \text{Max(Proj.i MW contrution on constraint)}}$$

6. The total NU costs for each project are calculated if more than one NU is required.

12. Revisions to the Draft report

- Table 2
 - M2 cost & cost pertaining to Interconnection Facilities which thus impacted the total cost and M3 amount.
 - Added a column reflecting details pertaining to TO network Upgrade (NU) cost self-

- funding
- Table 3
 - Provided costs of MTEP 21 Target A Project 20016¹
 - Added a column reflecting self-funding details
- Table 4
 - J1194, J1196, J1292, J1319 details updated in.
- Information pertaining to Table 2, Table 3 and Table 4 updated in Appendix A as well.

¹ <https://cdn.misoenergy.org/MTEP%20Projects%20Under%20Evaluation368757.xlsx>



Appendix A – Cost Allocation Summary (CEII)

Appendix B – Network Upgrade Table Per Project (CEII)

Appendix C – MISO ERIS Analysis (CEII)

Appendix D- MISO Stability Analysis

Appendix E – ITC Short Circuit Analysis (CEII)

Appendix F – PJM Affected Systems Study Report (CEII)

Appendix G – CE Affected Systems Study Report (CEII)

Appendix H – DTE Affected Systems Study Report (CEII)

Appendix I – WPSC Affected Systems Study Report (CEII)

Appendix J – MISO Deliverability Analysis (CEII)

Appendix K– MISO A10-Conditionality Analysis (CEII)